Illustration:

Let us now try to find more rational numbers say between $\frac{3}{7}$ and $\frac{4}{7}$ by the following visual explanation on the number line. If we get the multiples of the denominator of the equivalent rational numbers (the easy one will be to multiply by 10), then we can insert as many rational numbers as we want. We shall write $\frac{3}{7}$ as $\frac{30}{70}$ and $\frac{4}{7}$ as $\frac{40}{70}$ and see that there are 9 rational numbers between $\frac{3}{7}$ and $\frac{4}{7}$ as given in the number line below.

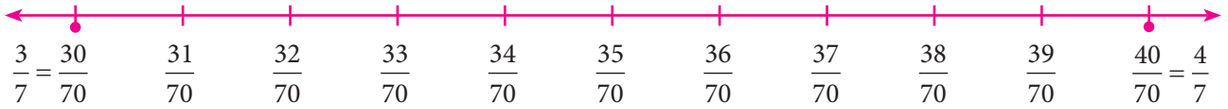


Fig. 1.12

Now, if we want more rational numbers between say $\frac{37}{70}$ and $\frac{38}{70}$ we can write $\frac{37}{70}$ as $\frac{370}{700}$ and $\frac{38}{70}$ as $\frac{380}{700}$. Then again, we will get nine rational numbers between $\frac{37}{70}$ and $\frac{38}{70}$ as $\frac{371}{700}, \frac{372}{700}, \frac{373}{700}, \frac{374}{700}, \frac{375}{700}, \frac{376}{700}, \frac{377}{700}, \frac{378}{700}$ and $\frac{379}{700}$.

The following diagram helps us to understand this nicely with a magnifying lens used between 0 and 1 and further zoomed into the fractional parts also.

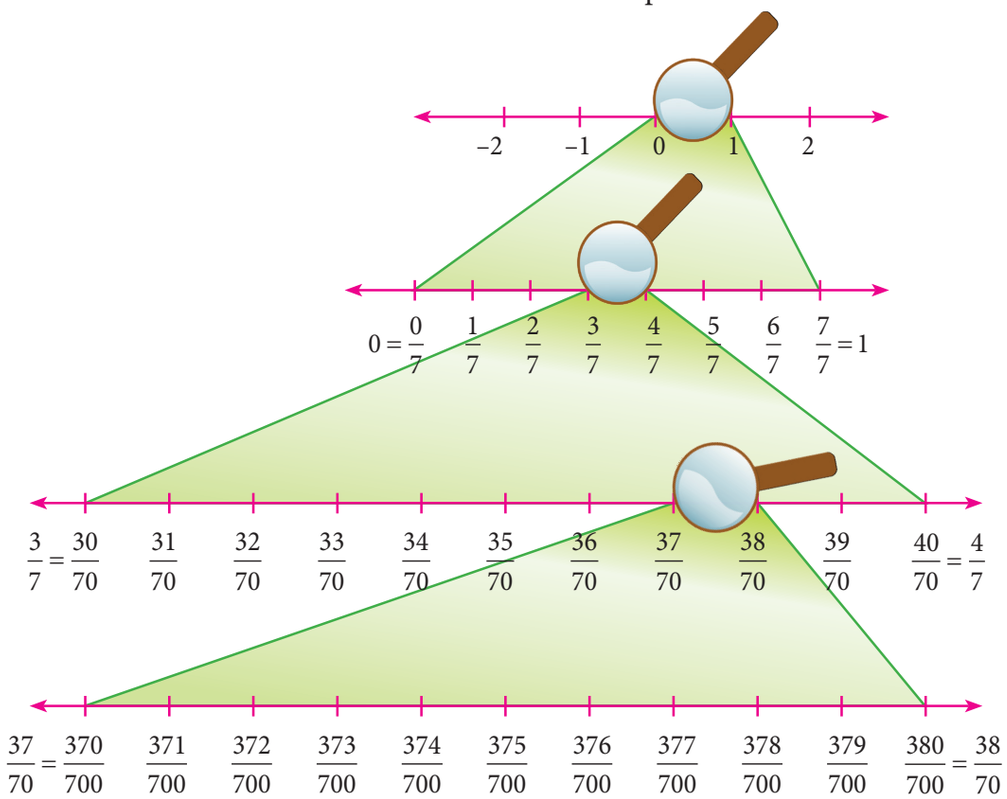


Fig. 1.13

Thus, we can see that there are an unlimited number of rational numbers between any two given rational numbers.

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